

HW7

Thomas Kidane

March 22, 2025

Definition 5.5.6

The random variables X and Y , defined on the same sample space, are said to be independent if for all x and y ,

$$P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y).$$

Lemma 5.5.9

The continuous random variables X and Y are independent if and only if they have a joint density $f(x, y)$ which can be written as a product $f(x, y) = g(x)h(y)$ of a function of x alone and a function of y alone. (We say, as before, that f factorises.)

Exercise 5.5.10

Prove this lemma. You may want to look at the proof of Theorem 2.4.8.

Claim 1($\rightarrow$): If $f(x, y)$ can be described as a function of x and y , then we can get set the distribution function of the two variables as such.

$$\begin{aligned} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy &= 1 \rightarrow \int_{-\infty}^{\infty} g(x) dx = 1, \int_{-\infty}^{\infty} h(y) dy = 1 \\ P(X \leq x) &= \int_{-\infty}^x g(l) \int_{-\infty}^{\infty} h(y) dy dl = \int_{-\infty}^x g(l) dl \int_{-\infty}^{\infty} h(y) dy = \int_{-\infty}^x g(x) dx \\ P(Y \leq y) &= \int_{-\infty}^y h(l) \int_{-\infty}^{\infty} g(x) dx dl = \int_{-\infty}^y h(l) dl \int_{-\infty}^{\infty} g(x) dx = \int_{-\infty}^y h(y) dy \end{aligned}$$

But $P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f(x, y) dx dy = \int_{-\infty}^x g(l) dl \int_{-\infty}^y h(k) dk = P(X \leq x)P(Y \leq y)$, therefore X and Y are independent.

Claim 2($\leftarrow$):

If X and Y are independent then $P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y)$. So we can make a function $g'(x) = P(X \leq x) \rightarrow g(x) = \frac{d}{dx}P(X \leq x)$, and same definition for $h(y)$.